Storytelling as Inverse Inverse Planning

Chandra, Li, Tenenbaum & Ragan-Kelley (2023), read with Borges, *The Garden of Forking Paths*

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Master-Seminar: Theory of Mind

21.01.27

Scan for notes and slides

Motivation

Two triangles and a circle

🎞 Heider & Simmel (1944)

A 90-second animation of simple shapes.

- viewers see agents with goals
- they make moral judgements
- no faces, no words, only motion

🤖 Can a computer do this?

First try: a rational planner.

- agents reach their goals
- but the motion is flat
- no suspense, surprise or irony

💡 Question of the paper

What does a storyteller do that a rational agent does not?

🔗 Animations as visual ToM tests: 19.11. (multimodal ToM benchmark)

Acting *in* vs. acting *out* a situation

Acting *in* a situation

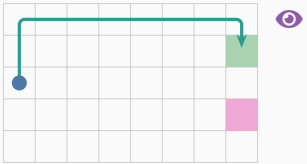
Optimize the character's reward



shortest path: 8 steps, goal unclear until the end

Acting *out* a situation

Optimize the audience's belief



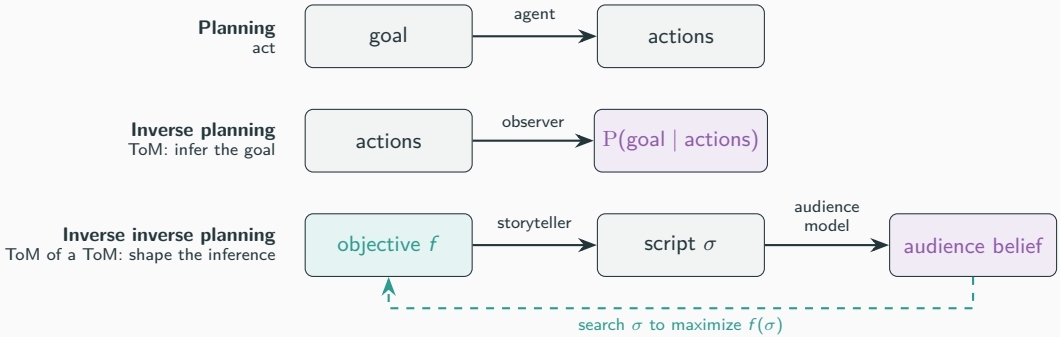
detour: 10 steps, the audience sees the goal at once

Stories as "probability design" (Kukkonen, 2014)

Good stories design the journey of the reader's beliefs.

Background

Planning, inverse planning, inverse inverse planning



Key idea

The storyteller plans *through* a model of the audience's theory of mind.

Bayesian inverse planning

Theory of mind as Bayes' rule

$$\mathrm{P}(\text{goal} \mid \text{action}) \propto \mathrm{P}(\text{action} \mid \text{goal})\,\mathrm{P}(\text{goal})$$

- likelihood: softmax-rational agent
- prior: plausible goals

Earlier in this seminar

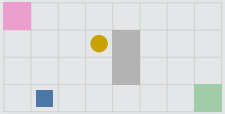
- 03.12.: Bayesian ToM (Baker et al.)
- 17.12.: LLM-augmented inverse planning
- 14.01.: `memo` hands-on

This paper

Inverse planning is not used to *read* minds, but as the audience model.

The audience model: help or hinder?

Grid-world kitchen



strong robot, weak cheese (moves succeed 60%),

goal tiles pink and green, a gray table

Social reward

- robot gets $\rho \cdot r_{\text{cheese}}$ extra
- $\rho > 0$ helps, $\rho < 0$ hinders
- robot models the cheese modelling the robot

Observer

$$\mathrm{P}(H \mid s \to a) \;\propto\; \mathrm{P}(s \to a \mid H)\,\mathrm{P}(H), \qquad H = \langle G_{\text{cheese}}, G_{\text{robot}}, \rho \rangle$$

Audience model from Ullman et al. (2009). Full hypothesis space in the appendix.

Inverse inverse planning

Optimizing a script

🎞 Script

$\sigma = (s_0,\ \sigma_1, \ldots, \sigma_T)$

$\sigma_t = \langle a_t^{\text{robot}}, a_t^{\text{cheese}}, s_t \rangle$

- the setting s_0 is optimized too
- so is the outcome of each move

🎯 Objective: depict "helping"

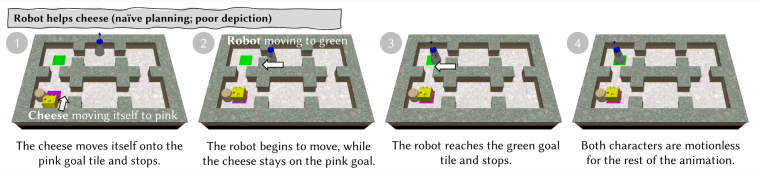
$$f_{\text{help}}(\sigma) = \sum_{t=1}^{T} \mathrm{P}(\rho_{\text{robot}} > 0 \mid \sigma_{1:t})$$

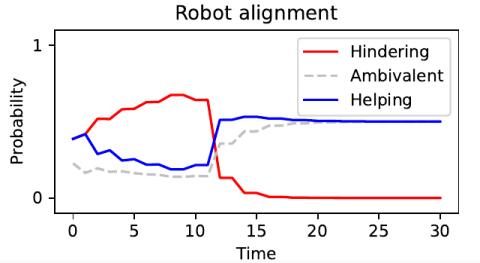
- strong belief at every step
- beam search over scripts

💡 Abstract objective

f_{help} fixes no goals or positions. The optimizer finds a scene that shows helping.

Naïve planning: the observer stays unsure



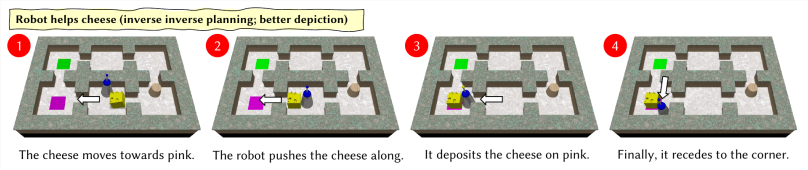


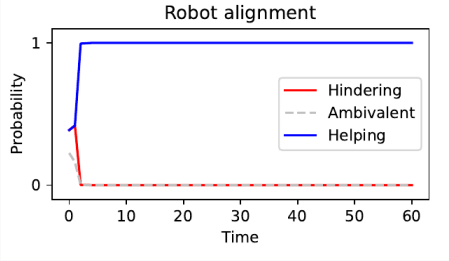
How to read the plot

- y: observer's posterior for the robot's alignment: hindering, ambivalent, helping
- x: time step of the animation
- helping and ambivalent share the probability: the robot only walks to green

Model output from Chandra et al. (2023), Fig. 1.

Inverse inverse planning: the observer is sure





Same axes as before

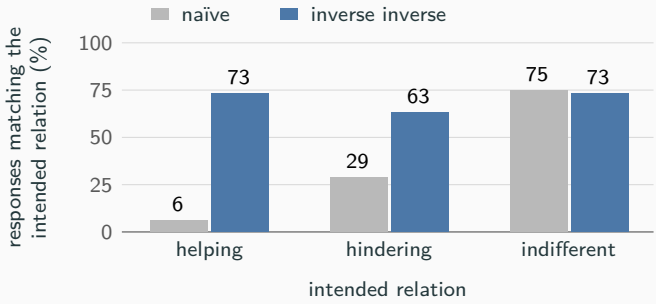
- the setting is optimized too: the robot pushes the cheese to pink, then steps back
- helping stays near 1 for almost the whole video
- f_{help} = area under the helping curve

Model output from Chandra et al. (2023), Fig. 2.

Behavioral experiment 1: do humans agree?

Setup

- 120 animations: 20 seeds × 3 relations × 2 methods
- 98 participants, 15 animations each
- answer: help / hinder / indifferent / unclear



Mean of 20 animations per bar. Help, hinder: $p < 0.01$. 95% CIs: paper Fig. 3.

Reading

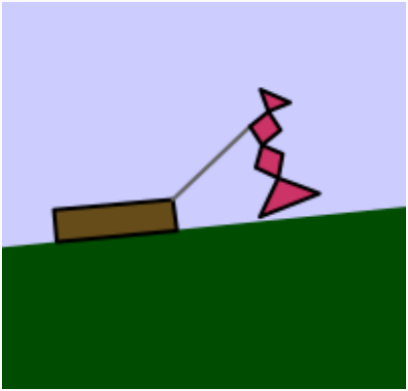
Indifferent characters rarely interact, so naïve planning shows indifference well. Helping is hardest: not all scenes allow it, for example when the cheese is already close to its goal.

Human vs. model comparison of ToM: 29.10.

Miming by “inverse inverse physics”

A hopper pulls a box uphill

- audience model: intuitive physics, not ToM
- the observer infers the box weight from the motion
- inverse inverse planning: make a light box look heavy
- the hopper “pretends to struggle”



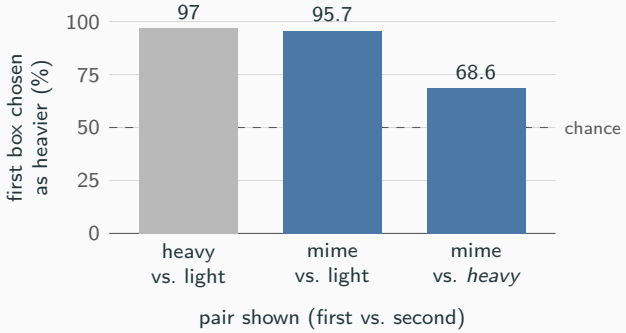
Chandra et al. (2023), Fig. 4a.

Behavioral experiment 2: is the mime convincing?

Setup

- 35 participants, 12 pairs each
- each video: honest light, honest heavy, or mime (light box)
- task: which box is heavier?
- not told that hoppers can mime

Fig. 4b caption says N = 70.



Same-condition pairs: chance. Heavy vs. light: $p < 0.01$.

95% CIs: paper Fig. 4b.

Reading

The mime looks heavier than a box that is heavier in 68.6% of choices.

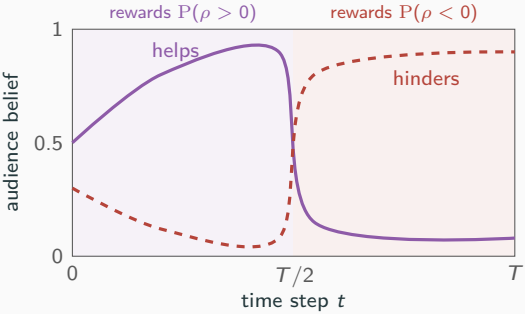
Storytelling device: plot twist

Objective

$$f_{\text{twist}} = \sum_t \begin{cases} \mathrm{P}(\rho > 0 \mid \sigma_{1:t}) & t \le T/2 \\ \mathrm{P}(\rho < 0 \mid \sigma_{1:t}) & t > T/2 \end{cases}$$

Generated scene

The robot "helps" the cheese to pink, then keeps pushing and traps it along the wall.



Schematic: the curve shape f_{twist} rewards, not model output.

Storytelling devices: irony and flashback

Dramatic irony

Audience knows what a character does not.

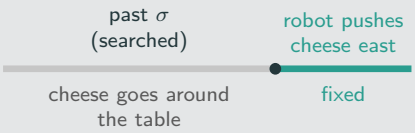
$$f_{\text{irony}} = \sum_t \Big(\mathrm{P}(G_c{=}g \mid \sigma_{1:t}) + \mathrm{P}(\rho < 0 \mid \sigma_{1:t}, G_c{=}g) + \mathrm{P}(\rho > 0 \mid \sigma_{1:t}, G_c{=}p) \Big)$$

Robot tries to help, but hinders: wrong belief about the goal.

Flashback

$$f_{\text{flashback}}(\sigma) = f_{\text{help}}\big(c(\sigma)\big)$$

Append a known event to σ. Search for a past that recasts it as helping.



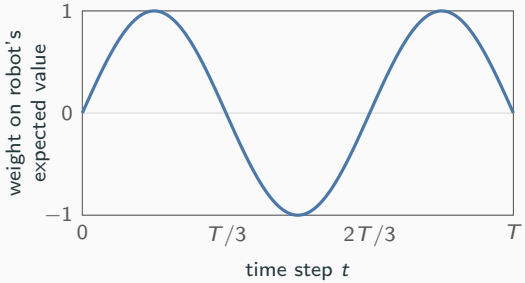
Irony needs a false belief in a character, as in false-belief tasks: 29.10.

Storytelling device: narrative arc

Objective

$$f_{\text{arc}} = \sum_t \sin\left(\tfrac{t}{T} 3\pi\right) \mathbb{E}\left[V^H_{\text{robot}}(s_t) \mid \sigma_{1:t}\right] - 0.1\, D_{\text{KL}}(H_{1:t-1} \,\|\, H_{1:t})$$

- rise, fall, rise: Vonnegut's story shapes
- KL penalty keeps apparent goals stable
- deus ex machina: a table falls onto pink; no character expects it



Exact weight $\sin(3\pi t/T)$ from f_{arc}. Positive: reward a high value. Negative: reward a low value.

All device examples: same random seed (0).

The Garden of Forking Paths

Borges (1941): the plot in one slide

The situation

- Yu Tsun, German spy in England, 1916
- knows the town of a British artillery park
- Captain Madden is hunting him
- the Chief in Berlin only reads newspapers

What he does

finds “Stephen Albert” in the phone book

↓

visits him: they talk about a labyrinth-novel

↓

shoots Albert, is arrested

↓

newspapers report the murder

↓

the Chief reads them: the town of Albert is bombed

Reading 1: the murder as inverse inverse planning



Acting *out*, not *in*

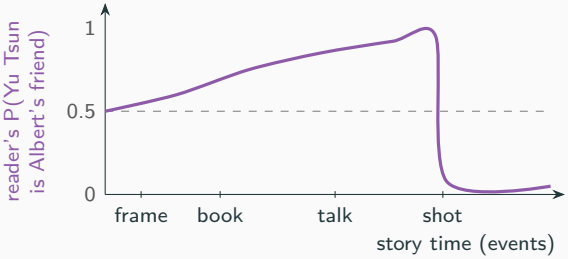
- in the world: kills a man he just met, gets hanged
- for the belief: the only action that works
- the Chief infers Yu Tsun's goal: ToM

The extreme mime

Like the hopper: a murder that *means* a place name.

Mapping = our reading of the story, not a claim of the paper.

Reading 2: Borges and the reader's beliefs



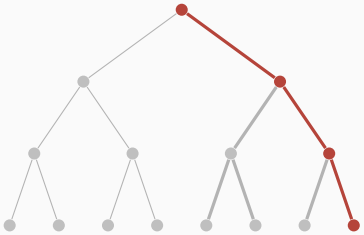
illustrative sketch, not data

🔗 Nested beliefs ("the Chief knows that I know . . . "): epistemic logic, 04.02.

🔀 Devices, applied to Borges

- **recasting**: the opening frame gives a known delay new meaning
- **twist**: "friend", then the shot
- **irony**: Albert says in some futures he is the enemy

Reading 3: the garden as the space of scripts



Ts'ui Pen takes *every* branch.

The storyteller keeps one σ.

The chess riddle

- Albert: the novel is about time, but never uses the word
- the reader must infer the concept
- Yu Tsun never sends the word "Albert"

Show, don't tell

Choose one path for its effect on the audience.

Discussion

The evidence: what does it show?

✔ Strengths

- humans judge the output, not the model's own score
- one method, two audience models: ToM and physics
- each device is a short objective you can read

🔍 Concerns (our critique)

- weak baseline: the naïve planner does not try to be understood
- devices: one seed each, no human test
- one small grid world, 20 seeds per relation
- Exp. 2: small sample, N given as 35 and as 70

⚖ Claim vs. evidence

Shown: the method *can* make scenes that people read as intended.
Not shown: that human storytellers plan this way.

A stronger baseline: legible motion planning (Dragan et al., 2013), which the paper cites.

Limits and critique

From the paper

- scales badly: state, hypothesis, script spaces
- too slow for real-time behavioral experiments
- future work: emotion in the audience model

Raised by Borges

- **shared hypotheses**: the Chief must consider "a killing encodes a name"
- **channel**: one lossy report, not a full trajectory
- **unreliable narrator**: the editor's footnote

Ethics (our extension)

The same machinery can deceive. Robotics already studies both legible and deceptive motion.

Audience aware of being modelled: 07.01. (interlocutor awareness) Recursive ToM: 10.12. invited talk



Questions

Discussion

1. The model makes readable scenes. What data would show that *people* tell stories this way?
2. Objective for Yu Tsun: which term makes the murder win?
3. Is the Chief a Bayesian inverse planner? What prior does he need?
4. Can the garden be optimized, or is refusing to choose the point?
5. The arc objective *penalizes* belief jumps (D_{KL}). Should a twist *reward* one?
6. Could an LLM be the audience model? How would we test it? 🔗 Clever Hans risk: 05.11.

Sources i


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Heider, F. and Simmel, M. (1944).
An experimental study of apparent behavior.
The American Journal of Psychology, 57(2):243–259.

Sources ii



Kukkonen, K. (2014).
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Anglia, 132(4):720–739.



Ullman, T., Baker, C., Macindoe, O., Evans, O., Goodman, N., and Tenenbaum, J. B. (2009).
Help or hinder: Bayesian models of social goal inference.
In *Advances in Neural Information Processing Systems*, volume 22.

Appendix: model details

Grid world

- multi-agent MDP, actions $\{\leftarrow, \rightarrow, \uparrow, \downarrow, \text{stay}\}$
- $H = \langle G_{\text{cheese}} \in \{p, g\}, G_{\text{robot}} \in \{p, g, \emptyset\}, \rho \in \{0, \pm 1, \pm 3\}\rangle$
- $\mathrm{P}(s \to a \mid H) \propto \exp\left(\beta\, Q_c^H(s, a)\right)$
- value iteration, beam search, $T = 15$

Physics and arc

- hopper: SHAC, policy $\pi(s; \theta)$, value $V(s; \phi)$
- box weights 0.1 (light), 0.5 (heavy)
- $f_{\text{arc}} = \sum_t \sin\left(\frac{t}{T} 3\pi\right) \mathbb{E}[V^H_{\text{robot}}(s_t) \mid \sigma_{1:t}] - 0.1\, D_{\mathrm{KL}}(H_{1:t-1} \| H_{1:t})$

Related work and code

Planning-based storytelling (Riedl & Young), RSA pragmatics (Frank & Goodman). The audience model could be written in `memo` (14.01.). Code and videos: `https://people.csail.mit.edu/kach/a2i2p/`